ISEP INSTITUTO SUPERIOR DE ENGENHARIA DO PORTO

DEPARTAMENTO DE ENGENHARIA INFORMÁTICA

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OS PICAS



RAILWAY NETWORK SIMULATOR

USER MANUAL

1240901, Bruno Monteiro
1241039, Francisco Vieira
1241478, Lourenço Aragão
1241129, Gustavo Ferreira

Oriented by: Ana Barata

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2. GLOSSARY

TEA (EN)	DESCRIPTION (EN)
AC	Acceptance Criteria
Acceleration	The rate at which a train increases its speed
Agile Methodology	A software development approach that emphasizes iterative progress, collaboration, and adaptability
Assertion	A condition used in automated tests to verify that the actual result matches the expected result.A statement in a unit test that checks if a specific condition is true, used to validate the behavior of code
Authentication	The process of confirming a user's identity, usually through a username and password, to ensure secure access to the system and its features
Automobile	A type of final product cargo produced by transforming industries such as car factories. Transported between stations for delivery to cities
Backlog	A prioritized list of tasks, features, or user stories that need to be completed in a project
Boxplot	Graphical representation of data distribution using quartiles
Budget	Financial resources available to the player for acquiring locomotives, building stations, and railway lines
Building	A structure with walls and a roof, such as a house, factory, or station building, used within the game environment
BuildingType	Enumeration that defines the types of buildings that can be used to upgrade stations
Café	A station facility that boosts passenger satisfaction by offering refreshments. Comes in small and large variants with varying service levels
CamelCase	Variable naming convention where the first word is always in lowercase and every other word starts with a capital letter
Cargo	Goods that can be transported by trains, including passengers, mail, raw materials, and finished products
CargoType	Enumeration that defines the different types of cargo (e.g., passengers, mail, coal, cars)

Carriage	Non-powered rail vehicle used to carry specific cargo types, such as passengers, mail, coal, or cars
Cattle	A raw resource generated by farms in the primary sector, typically used in transforming industries to produce food or other processed goods
City	Populated area on the map with a name, coordinates and housing blocks
Coal	A natural resource obtained from mines, serving as a key raw material for transforming industries such as steel production
Coordinate	A system that defines the exact position of elements on the map using X and Y values
Custom	A station upgrade that manages and regulates international cargo, typically added near ports or border stations to handle cross-border trade
DTO	Acronym for *Data Transfer Object*
Data Transfer Object	An object used to carry data between layers of the application, minimizing direct coupling with domain entities
Depot	Smallest type of station, with limited economic radius (3x3), lower cost (50k)
Diesel	A type of fuel used to power diesel locomotives, enabling train movement on non-electrified railway lines
Domain Model	A conceptual representation of real-world entities and their relationships within the problem domain, used to design and structure the software
Economic Radius	Area around a station where industries supply or demand cargo
Editor	Person who can create maps and scenarios that can be provided in a bundle with the game or sold separately
Electrified Line	Railway line that allows operation of electric locomotives
FXML	A markup language used with JavaFX to declaratively define the layout of user interfaces
Final Product	Goods such as food, textiles, and automobiles that are produced by transforming industries and consumed by cities
Fleury Algorithm	An algorithm used to find Eulerian paths in the railway network, necessary for line maintenance features.
Freight Train	Railway train for carrying goods rather than people

Game Simulation	An interactive digital model of railway operations where players manage stations, trains, routes, and cargo flow according to defined rules
GitHub	Web-based platform for version control and collaboration, using Git, that enables developers to host, manage, and track changes to code repositories
Git	Version control system designed to track changes in source code during software development
Graph	A mathematical structure used to represent the railway network topology (stations as vertices, railway lines as edges)
Graphviz	A graph visualization tool used to visually represent the railway network
Graphical Interface	A user interface based on visual and interactive elements such as buttons, menus, and windows, implemented using JavaFX
Histogram	A chart that plots the distribution of a numeric variable's values as a series of bars
Hotel	A station upgrade that provides enhanced passenger services. Available in small and large sizes, improving passenger satisfaction and station capacity
House Block	Representation of residential areas within a city on the map
Incremental Development	A software development approach where the project is built and improved in small, manageable steps, adding functionality gradually
Industry	Facility that generates or processes resources in the simulation
Infrastructure	The physical and digital components of a railway system, including tracks, stations, trains, and supporting technology
Iron	A natural raw material mined from deposits, used along with coal to produce steel in transforming industries like steel mills
Jacoco	An open-source framework that measures how well automated tests exercise Java code
Java	Primary language for developing the project
Javadoc	Tool for generating useful documentation for Java code
JavaFX	A software platform used to build the graphical user interface (GUI) for the simulator
JUnit5	Testing framework for Java applications, designed to write and run repeatable unit tests
Jupyter Notebook	Development environment of statistical analysis in Python

KPI	Key Performance Indicator
LaTex	Markup language for documenting theoretical concepts and analyses
Liquid Storage	A station facility designed to store and preserve liquid cargo, improving handling efficiency and extending cargo lifespan
Locomotive	Powered railway vehicle used for pulling trains
LocomotiveType	Enumeration that defines the types of locomotives available in the game, such as steam, diesel, and electric
Logbook	A tool for recording development progress, decisions, and important events during a project
Mail	A type of cargo generated and consumed by cities, representing written communication between locations
Manager	An individual or role tasked with supervising and managing operations, resources, and teams within the railway system or project
Map	A grid structure representing the world where cities, industries, and tracks are placed
Map Editor	A tool used to create rectangular maps by placing static elements such as cities and industries, enabling custom scenarios and gameplay environments
Mixed industry	Ports potentially have the capacity, depending on what is defined in the scenario, to import or export any cargo
MVP	Acronym for *Minimal Viable Product*
Minimal Viable Product	A simplified initial version of a software with core features that allows testing and validation before further development
Non-Funcional Requirement	System specifications that define how the system operates, covering aspects like security, performance, usability, and maintainability
Object serialization	Technique used to guarantee the persistence of the data in two consecutive runs
Overbuilding	The act of constructing new infrastructure or buildings excessively or unnecessarily, often leading to resource waste or inefficiency
Pie chart	Way of summarizing a set of nominal data or displaying the different values of a given variable
Player	Person who wants to play a game
Position	Object that represents the (X, Y) coordinates of elements on the map, such as cities, industries, and stations

Port	Industry that can import/export and process various types of cargo depending on the scenario
Post Office	A station building upgrade that enhances mail handling efficiency, increasing the volume and speed of mail transported through the station
Primary sector industry	Generate resources, such as mines and farms
Product Owner	The person representing the organization that intends to develop the game
Railroad	Land transportation infrastructure consisting of two parallel steel rails on which trains travel to carry passengers and cargo
RailRoad Tycoon	Trains game that covers the areas of gameplay, economy and engineering
Railway	Efficient, safe, and environmentally friendly way of moving people and goods
Repository (Java)	A class responsible for storing, accessing, and manipulating domain objects, typically following the Repository design pattern
Railway line	Connection between two train stations
Route	List of stations where the train passes
Rubber	A raw material produced by specialized farms in the primary sector, used in the manufacturing of goods such as automobiles
Scenario	Set of rules and restrictions (time, historical, technological) defined for a given map
Scenario Editor	Tool for designing and modifying structured sequences in games, simulations or interactive apps
Serialization	The process of converting objects into a persistent format (e.g., `.ser` files) to ensure continuity between application runs.
SCRUM	An agile framework for project management where teams work in short, iterative cycles called sprints to deliver incremental value
SCRUM Master	A team role in the SCRUM framework responsible for facilitating the process, removing obstacles, and helping the team follow Agile practices effectively
Silo	A storage facility at a station used to hold bulk goods like grain or coal, increasing station capacity and improving cargo handling efficiency
Simulator	System that manages the dynamics of train operations, cargo flow, and network performance
Sprint	A short, fixed-length period in SCRUM during which a team works to complete a specific set of tasks or deliverables

Station	Building that serves cities and industries; can be Depot, Station, or Terminal, each with different radius and cost
StationType	Enumeration that defines the types of stations available in the game, such as Depot, Station, and Terminal
Steel Mill	Branch of metallurgy dedicated to the manufacture and treatment of steels and cast irons
SVG	Acronym for *Scalable Vector Graphics*
TEA	Acronym for *Terms, Expressions and Acronyms*
Terminal	Largest and most expensive station type (5x5 radius, 200k cost)
Test-Driven Development	A software development approach in which tests are written before the code to be tested
Telegraph	A historical communication system used at train stations to send messages quickly, later replaced by the telephone
Telephone	A communication device that replaced the telegraph, improving coordination and information exchange at train stations
Textile	A final product made in transforming industries like textile factories, consumed by cities as part of their goods demand
Time Restriction	The period in which the simulation will take place
Train	Composition of locomotives and carriages used for transportation
Transforming industry	Transform resources provided by the Primary Sector into new products
US	Acronym for *User Story*
User Story	Short, simple description of a feature told from the perspective of the person who desires the new functionality
Unit Test	Software testing where individual components of code, such as functions or methods, are validated in isolation to ensure they behave as intended
Version Control	A system that tracks and manages changes to software code, enabling collaboration and history tracking across development teams
Wool	A raw material produced by sheep farms, commonly used as input for textile manufacturing in transforming industries

3. INTRODUCTION

This user manual is designed to help users effectively interact with the Railway Network Simulator, a software product developed under the Integrative Project (LAPR2). It provides clear and structured guidance to enable users to access, understand, and operate all features without requiring direct assistance from the development team.

The purpose of this guide is to ensure that both technical and non-technical users can independently use the simulator's tools. The manual contains detailed descriptions, visual aids, and instructions that allow users to create maps, configure scenarios, build and manage railway stations, purchase locomotives, define cargo routes, and operate the simulation.

The Railway Network Simulator is a modular desktop application that allows users to create and simulate simplified railway networks. It supports the placement of cities and industries on a customizable map, construction of railway stations and lines, and real-time simulation of cargo transport based on supply and demand principles.

Target Users

This manual is addressed to two main types of users:

- Editors: Users responsible for creating maps and configuring scenarios (e.g., choosing the time period, available locomotives, and defining historical restrictions).
- Players: Users responsible for interacting with the system by building infrastructure, managing trains and cargoes, and running the simulation to optimize performance.

4. SYSTEM OVERVIEW

4.1 Purpose and Use

The Railway Network Simulator is a modular software system developed to simulate the creation and operation of simplified railway networks. The platform provides editors and players with tools to design maps, configure scenarios, build railway infrastructure, manage trains, and run logistics simulations.

It can be used for educational, analytical, or entertainment purposes, enabling users to explore transport planning, cargo flow optimization, and infrastructure expansion strategies in a controlled virtual environment.

4.2 System Structure and Features

The system is structured into two core components and a set of analytical modules, each with specific functionalities:

1. Map and Scenario Editor

This component enables Editors to:

- **Create maps:** Rectangular grids with customizable dimensions and scale.
- **Place elements:** Cities (with house blocks) and industries (primary, transforming, mixed).
- **Configure scenarios:** Set time periods, available locomotive types (steam, diesel, electric), historical events (e.g., wars, vaccination campaigns), and cargo generation modifiers.

2. Simulation Tool

This component is used by Players to:

- **Build stations:** Choose from Depot, Station, or Terminal, each with unique radius and cost.
- **Upgrade stations:** Add buildings (e.g., telegraph, cafe, post office, hotel) to enhance services or capacity.
- **Construct railway lines:** Connect stations using different track types (single/double, electrified/non-electrified).
- **Buy locomotives:** Based on scenario year and fuel compatibility.
- **Assign trains to routes:** Define paths and loading strategies (FULL, HALF, AVAILABLE).
- **Run the simulation:** Trigger cargo generation, train movement, and dynamic demand updates.
- **Track financial results:** Observe income from deliveries and calculate profitability.

3. Analytical Modules

Used for technical analysis and simulation optimization:

- **Connectivity Analysis:** Check whether a certain type of locomotive (steam, diesel, electric) can travel between two selected stations, considering electrification and station type.
- **Maintenance Planning:** Generate a route that traverses each railway line exactly once (Eulerian path), for simulated maintenance scheduling.
- **Statistical Tools (Python):** Perform in-depth performance analysis using Jupyter Notebook:
  - Boxplots for passenger arrivals
  - Histograms for cargo delivery distribution
  - Pie charts for train and mail allocation
  - Linear regression for revenue prediction based on cargo quantity

4.3 Technologies and Dependencies

The Railway Network Simulator is developed in Java, with the following technological stack:

- **JavaFX**: Used to build the graphical user interface for simulation control and data visualization.
- **Graphviz:** Generates visual graphs of the railway network (e.g., connection topology).
- **JUnit5 & JaCoCo:** Used to perform unit testing and measure code coverage.
- **Object serialization:** Enables persistent saving and loading of maps, scenarios, and simulations.
- **Python + Jupyter Notebook:** Used for statistical analysis and visualization of simulation data during later development sprints.

4.4 Product Structure Diagram

The following diagram provides a high-level overview of the internal structure and module organization of the Railway Network Simulator. It illustrates how the main components interact, from map and scenario creation to simulation execution and analytical tools.

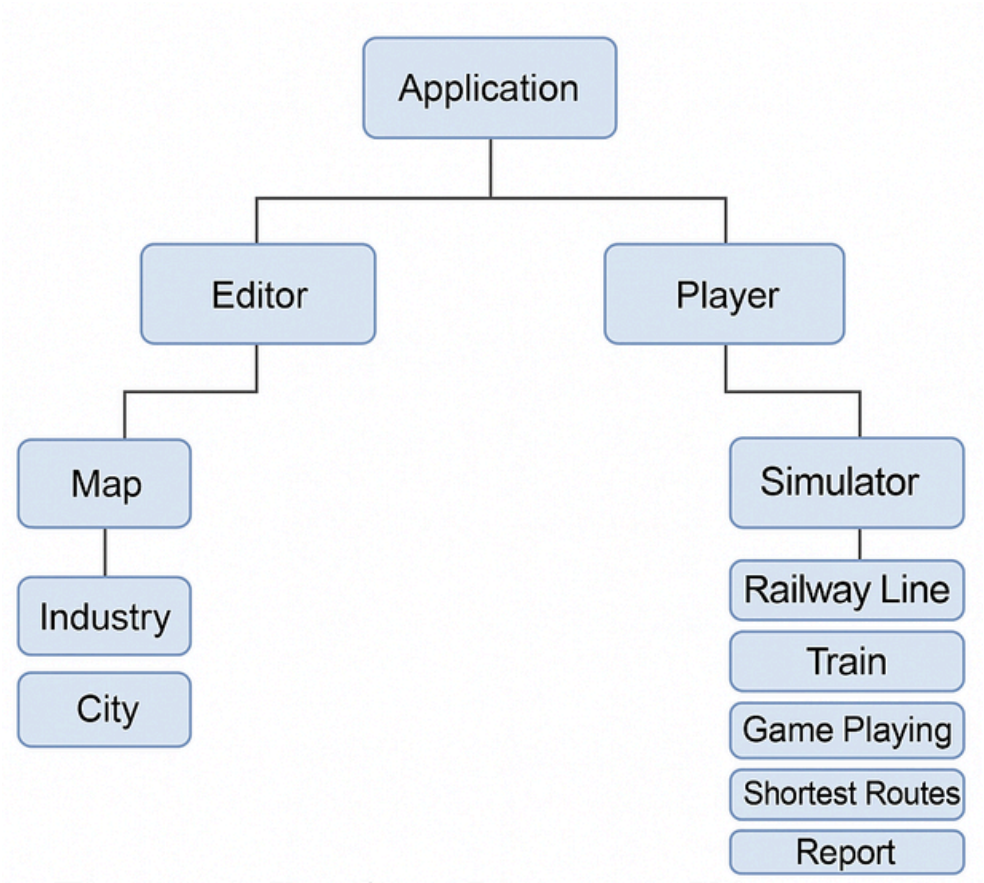


Figure 1 - Product Structure Diagram

5. SYSTEM REQUIREMENTS

To successfully install and run the Railway Network Simulator, the following hardware and software requirements must be met:

5.1 Operating System (OS)

- Windows 10 / 11 (64-bit)
- macOS 11.0 (Big Sur) or later
- Linux (Ubuntu 20.04 LTS or later)

5.2 Hardware Requirements

Minimum:

- RAM: 4 GB
- Processor: Dual-core 1.8 GHz
- Disk Space: 500 MB free

Recommended:

- RAM: 8 GB or higher
- Processor: Quad-core 2.5 GHz or higher
- Disk Space: 1 GB free

5.3 Software Requirements

- Java Development Kit (JDK) version 11 or higher
- JavaFX SDK version 11.0.2 or higher
- Graphviz (optional, for graphical network visualization)
- Python 3.x (for statistical analysis)
- Jupyter Notebook (for statistical modules in Sprint 2 and 3)
- Gnuplot: Used in Sprint 3 (US29) to visualize asymptotic behavior of algorithm execution times based on input size.

6. FEATURES

6.1 Member Login

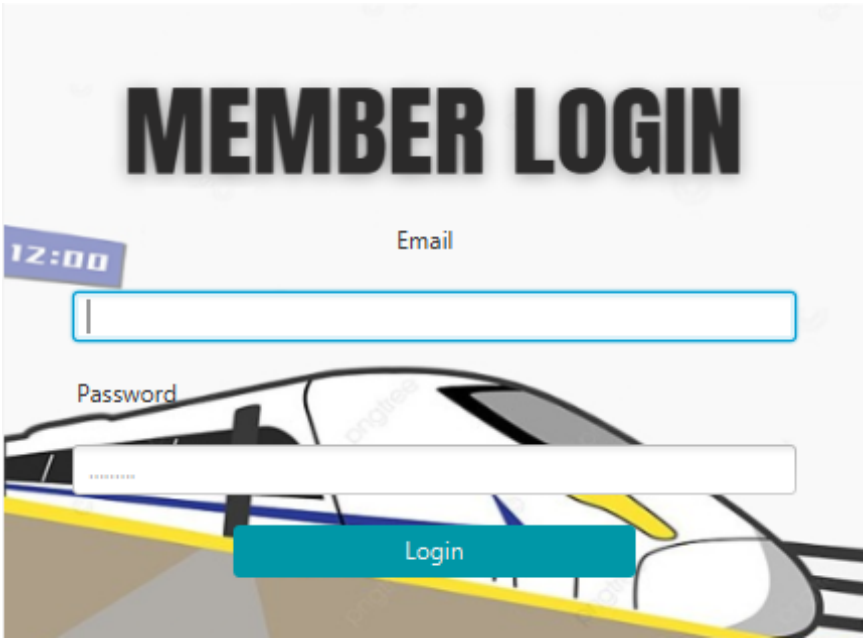


Figure 2 - Member Login

6.1.1 Editor Login

Enables the Editor to access all map and scenario editing functionalities.

Steps:

1. Open the application.
2. Enter the editor email(**editor@this.app**).
3. Enter the editor password(**editor**).
4. Click "Login".
5. Upon successful authentication, the system redirects to the Editor dashboard.

6.1.2 Player Login

Allows access to simulation-related functionalities.

Steps:

1. Open the application.
2. Enter the player email(**player@this.app**).
3. Enter the editor password(**player**).
4. Click "Login".
5. After successful login, the Player gains access to station, train, and simulation tools.

6.2 Editor Menu



Figure 3 - Editor Menu

After a successful login as an Editor, the user is presented with the **Editor Menu** screen. This interface provides direct access to all core functionalities needed to configure the simulation environment. The menu includes options such as:

- **Create Map**, which opens a form for defining a new map, including its name, dimensions, and scale.
- **Add Industry**, which allows the user to place different types of industries on the map at chosen positions.
- **Create City**, enabling the addition of a city with a specific number of housing blocks, either manually or automatically placed.
- **Create Scenario**, where the Editor can configure the operational rules of the simulation, such as the time period, available locomotives, and industry generation behavior.
- **Save Map**, which stores the current map configuration in a file for later reuse.
- **Load Map**, which loads a previously saved map file into the editor.
- **Save Scenario**, which saves the scenario's configuration and associated data.
- **Load Scenario**, which allows the Editor to continue working on a previously defined scenario.
- **Back to Login**, which returns the user to the authentication screen.

This centralized menu ensures intuitive navigation and efficient access to all Editor-specific features.

6.2.1. Create Map

If the Editor clicks on the "**Create Map**" it will enter the menu that allows creation of a rectangular map with defined size and scale.



Figure 4 - Create Map Menu

Steps:

1. Enter the name of the map.
2. Set width and height in number of cells.
3. Define the scale (e.g., 1 cell = 10 km).
4. Click "**Create Map**" to confirm creation.

Validation Rules

Before creating a map, the following validation rules must be respected:

- **Map Name:**
  - Must not be empty.
  - Must not contain special characters such as @, #, *, or /.
  - Only letters, numbers, and spaces are allowed.
- **Width and Height:**
  - Minimum value: 50 cells
  - Maximum value: (define if applicable)
  - Only positive integer values are accepted.
- **Scale:**
  - Must be a positive number.
  - Recommended range: 1 to 50 (e.g., 1 cell = 10 km).
  - No text or special characters allowed.

If any of the above conditions are not met, an error message will be displayed and the map will not be created.

6.2.2. Create Industry

Upon selecting "**Add Industry**", the user can insert a new industry by specifying its type and location on the map grid.

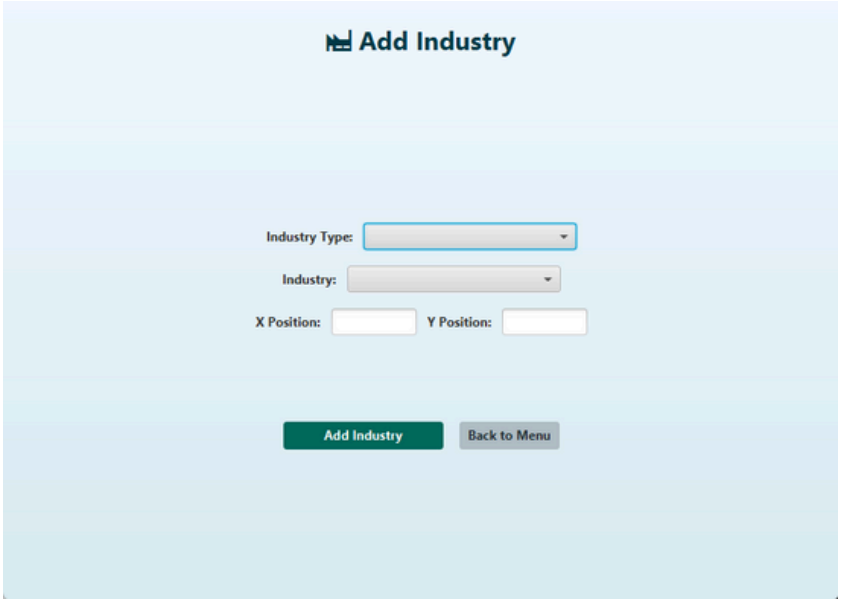


Figure 5 - Add Industry Menu

Steps:

1. Select the type of industry from the dropdown list(**Primary**, **Tranforming** or **Mixed**).
2. Chose the specific industry to add.
3. Enter the **X and Y coordinates** for placement.
4. Click "**Add Industry**" to confirm and place the industry on the map.

Validation Rules

Before adding an industry, the following validation rules must be respected:

- **Coordinates (X, Y):**
  - Must be inside the map boundaries (i.e., 0 ≤ X < width and 0 ≤ Y < height of the map).
  - Cannot overlap with another existing city or industry.

If any of the above conditions are not met, an error message will be shown and the industry will not be placed on the map.

6.2.3. Create City

When the Editor selects "**Create City**", the system opens a menu where a new city can be added to the map.

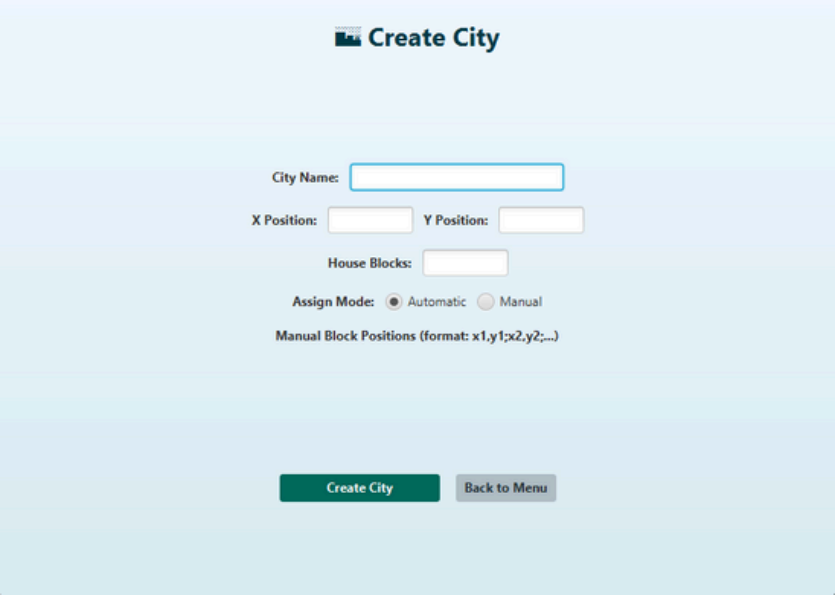


Figure 6 - Create City Menu (Automatic)

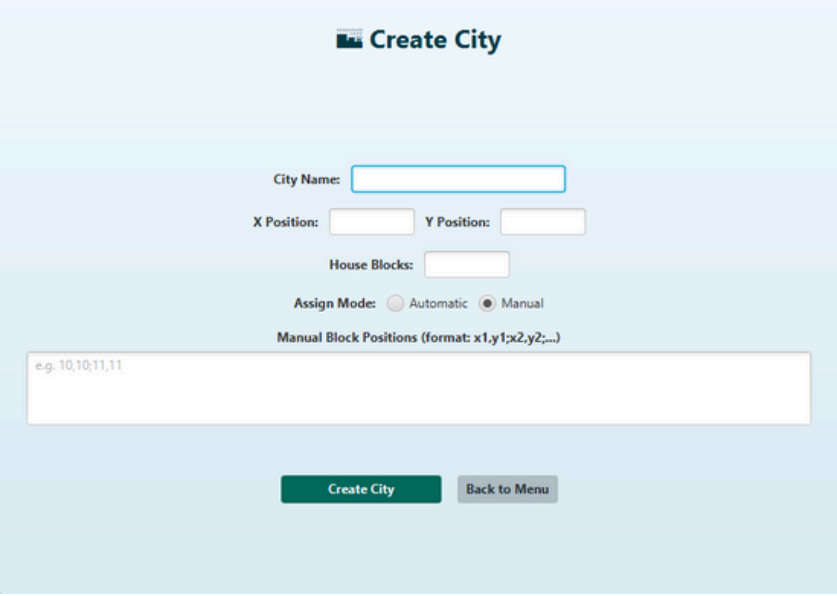


Figure 7 - Create City Menu (Manual)

Steps:

1. Enter the city's name.
2. Enter the **X and Y coordinates** for placement.
3. Define the number of house blocks.
4. Choose the placement method:
   - By default, **Automatic** is selected, and the system generates block positions around the city center.
   - Optionally, switch to **Manual** to manually insert the coordinates of each block.
5. Confirm to add the city to the map.

Validation Rules

Before creating a map, the following validation rules must be respected:

- **City Name:**
  - Cannot be empty or null.
  - Must not contain special characters such as @, #, *, or /.
  - Only letters, numbers, and spaces are allowed.
- **Coordinates (X, Y):**
  - Must be inside the map boundaries (i.e., 0 ≤ X < width and 0 ≤ Y < height of the map).
  - Cannot overlap with another existing city or industry.
- **Number of House Blocks:**
  - Must be a positive number (Minimum: 1 block).
- **Block Coordinates (Manual Mode only):**
  - Each coordinate (X, Y) must be within map boundaries.
  - No duplicate or overlapping positions.

If any of the above conditions are not respected, the system will display an error and prevent the creation of the city.

6.2.4. Create Scenario

When the Editor selects "**Create Scenario**", the system opens a configuration menu where they can define the rules and restrictions that apply to a specific map during simulation.

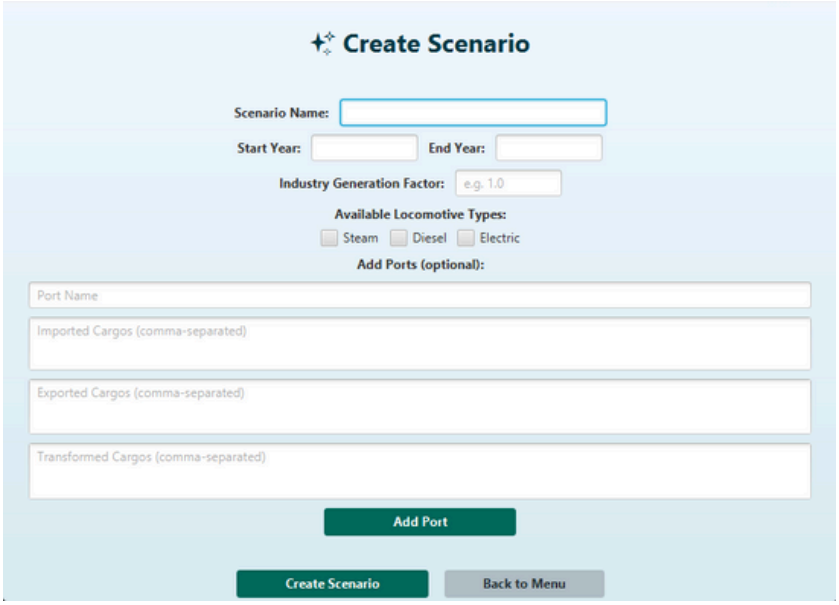


Figure 8 - Create Scenario

Steps:

1. Enter the scenario name in the "**Scenario Name**" field.
2. Insert the time range by filling in the **Start Year** and **End Year**. These define the operational period of the scenario.
3. Set the **Industry Generation Factor** in the appropriate field. This value (e.g., 1.0) determines how many new industries are created per year.
4. Select the allowed locomotive types by ticking one or more checkboxes (**Steam**, **Diesel** and **Eletric**)
5. *(Optional)* To define ports, scroll to the "Add Ports (optional)" section:
   - Enter the **Port Name**.
   - List the **Imported Cargos**, separated by commas (e.g., coal, mail, rubber).
   - List the **Exported Cargos**, separated by commas.
   - List the **Transformed Cargos**, separated by commas.
   - Click "**Add Port**" to include it in the scenario.
6. After configuring all fields, click "**Create Scenario**" to confirm and save.

Validation Rules

Before creating a scenario, the following validation rules must be respected:

- **Scenario Name:**
  - Cannot be empty or null.
  - Must not contain special characters such as @, #, *, or /.
  - Must be unique (no duplicates allowed).
- **Start Year and End Year:**
  - Both fields must be filled
  - Must be valid integers.
  - Start Year must be less than End Year
- **Industry Generation Factor:**
  - Must be a positive decimal or integer (e.g., 1.0, 2, etc.)
- **Locomotive Types:**
  - At least one option (Steam, Diesel, Electric) must be selected
- **Ports (optional):**
  - **Port Name**: cannot be empty or duplicated
  - **Imported/Exported/Transformed Cargos**:
    - If filled, must be a list of valid cargo types, separated by commas.
    - Cargo names must not contain numbers or special characters.

If any of the above conditions are not respected, the system will display an error and prevent the creation of the city.

6.2.5. Save Map

When the Editor selects "**Save Map**", the system allows the current map to be stored as a file for future use.

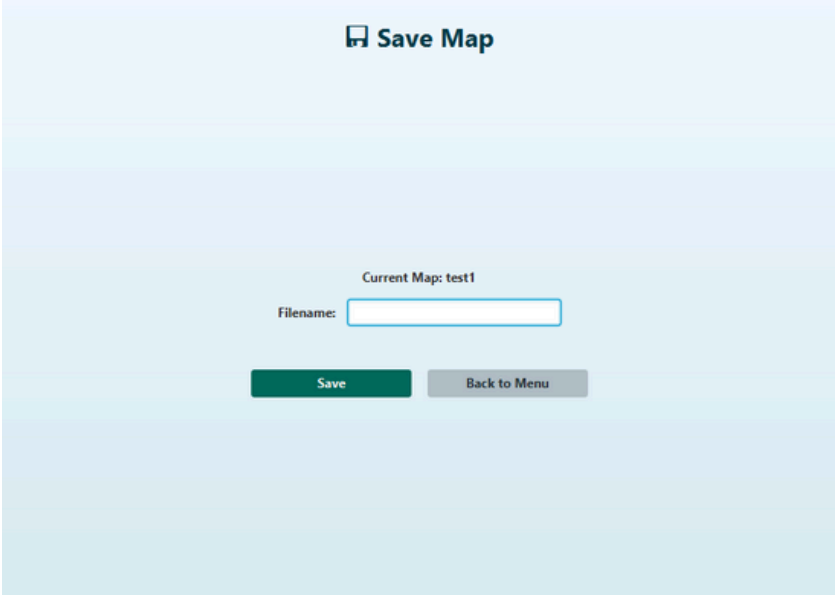


Figure 9 - Save Map Menu

Steps:

1. Click on the "**Save Map**" option in the Editor Menu.
2. Enter a valid name for the map file.
3. Confirm the save operation.
4. The map is saved to a .ser file, ensuring persistence between sessions.

Validation Rules

Before saving a map, the following conditions must be respected:

- **Filename:**
  - Cannot be empty or null.
  - Must not contain special characters.
  - Must be unique (i.e., not overwrite another file unless confirmed by the user).
- **Map Name:**
  - A valid map must be selected (i.e., an active map must exist).

If any of the above conditions are not respected, the system will display an error and prevent the creation of the city.

6.2.6. Load Map

When the Editor selects "**Load Map**", the system allows them to load a previously saved map from a file.

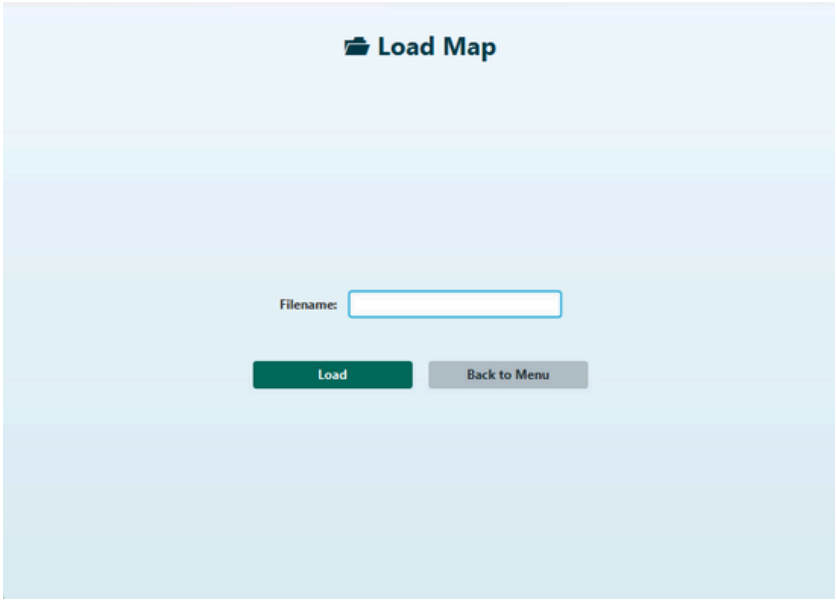


Figure 10 - Load Map Menu

Steps:

1. Click on the "**Load Map**" option in the Editor Menu.
2. Enter the name of the previously saved map file in the **Filename** field.
3. Click "**Load**" to import the map and make it the active one.
4. The map becomes available for use in city, industry, and scenario creation.

Validation Rules

Before saving a map, the following conditions must be respected:

- **Filename:**
  - Must correspond to an existing saved map file.

If any of these conditions are not met, the system will prevent the operation and inform the user via an error dialog.

6.2.7. Save Scenario

When the Editor selects "**Save Scenario**", the system allows them to save the currently active scenario to a file.



Figure 11 - Save Scenario Menu

Steps:

1. Click on the "**Save Scenario**" option in the Editor Menu.
2. The current scenario name is displayed (read-only).
3. Enter a filename to save the scenario.
4. Click "**Save**" to confirm and store the scenario configuration.

Validation Rules

Before saving a map, the following conditions must be respected:

- **Filename:**
  - Cannot be empty or null.
  - Must not contain special characters.
  - Must be unique (i.e., not overwrite another file unless confirmed by the user).
- **Active Scenario:**
  - A scenario must be active in the session.
  - The system will not allow saving if no scenario has been created or loaded.

If any of these conditions are not met, the system will block the save operation and display a warning or error message.

6.2.8. Load Scenario

When the Editor selects "**Load Scenario**", the system allows them to retrieve a previously saved scenario by specifying the corresponding filename. This enables the user to resume editing or simulation preparation.

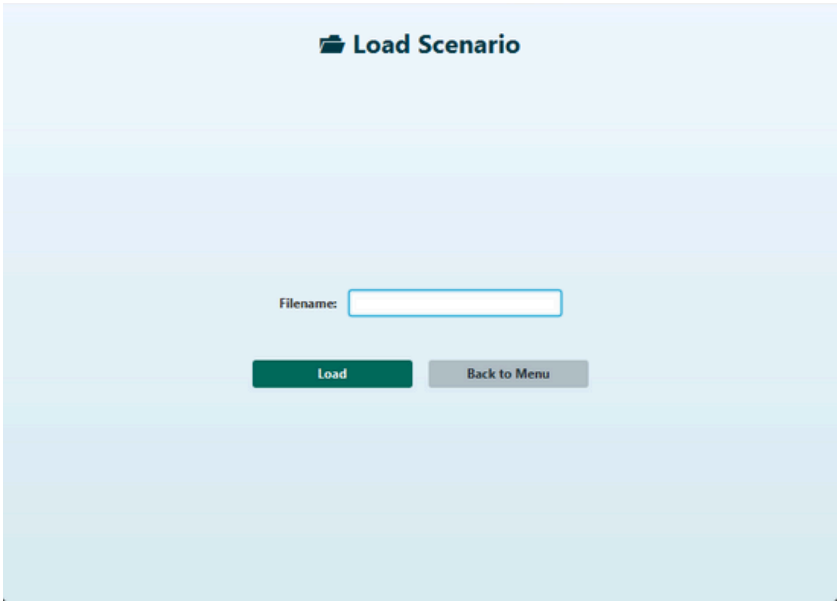


Figure 12 - Load Scenario Menu

Steps:

1. Click on the "**Load Scenario**" option in the Editor Menu.
2. Enter the name of a previously saved scenario file in the **Filename** field.
3. Click "**Load**" to open the file..
4. The scenario becomes active and can be edited or used in simulation.

Validation Rules

Before a scenario can be loaded, the following conditions must be respected:

- **Filename:**
  - Must correspond to an existing saved map file.

If any of the above conditions are not respected, the system will display an error and prevent the creation of the city.

6.3 Player Menu

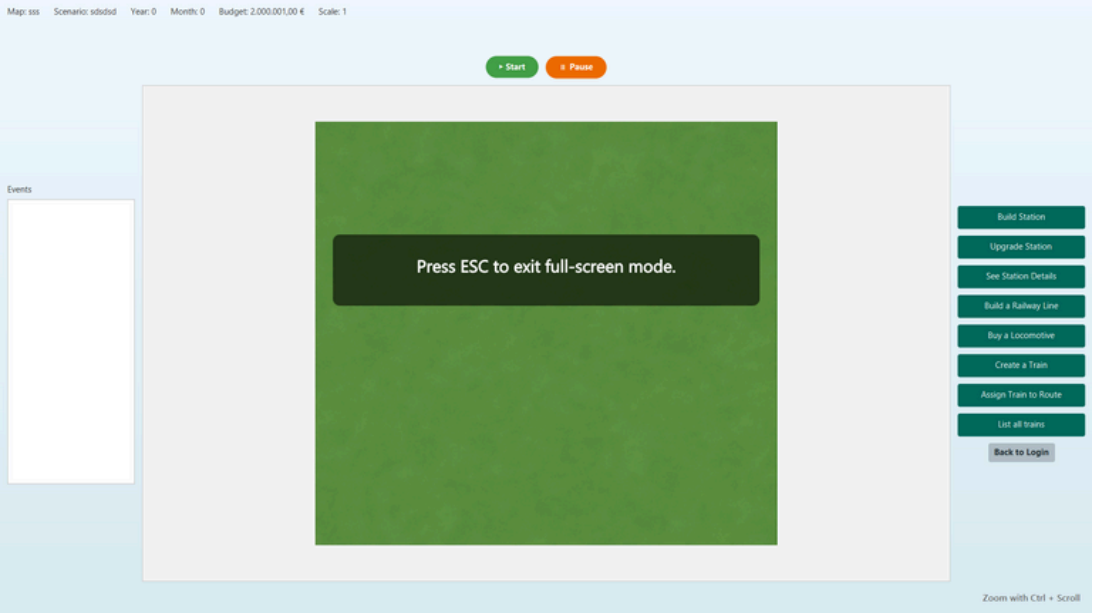


Figure 13 - Player Menu

After a successful login as a Player, the user is presented with the Player Menu screen. This interface provides direct access to all core functionalities required to operate the railway simulation. The menu includes options such as:

- **Build Station**, which allows the player to construct a new station (Depot, Station, or Terminal) on a selected position of the map.
- **Upgrade Station**, where the player can install buildings (e.g., telegraph, café, hotel) to increase the station's capacity and service quality.
- **See Station Details**, which displays detailed information about a selected station, including its buildings, connected industries, cargo types, and economic radius.
- **Build a Railway Line**, which enables the player to connect two stations with a railway line (electrified or not), enabling train travel between them.
- **Buy a Locomotive**, which enables the purchase of locomotives available according to the scenario's year and allowed fuel types.
- **Create a Train**, which allows combining a locomotive with cargo units to form a train that can be assigned to a route.
- **Assign Train to Route**, which allows defining a route and loading strategy (FULL, HALF, AVAILABLE) for a selected train.
- **List all trains**, which shows a list of all created trains along with their routes and current status.
- **Back to Login**, which returns the user to the authentication screen.

At the top of the screen, the user can also Start or Pause the simulation, while viewing contextual information such as Map Name, Scenario, Year, Budget, and Scale. On the left side, an Events panel is reserved for simulation messages or alerts.
This centralized menu ensures intuitive navigation and efficient access to all Player-specific features.

6.3.1. Build Station

If the Player clicks on the "**Build Station**" option, it will enter the menu that allows the construction of a new railway station.

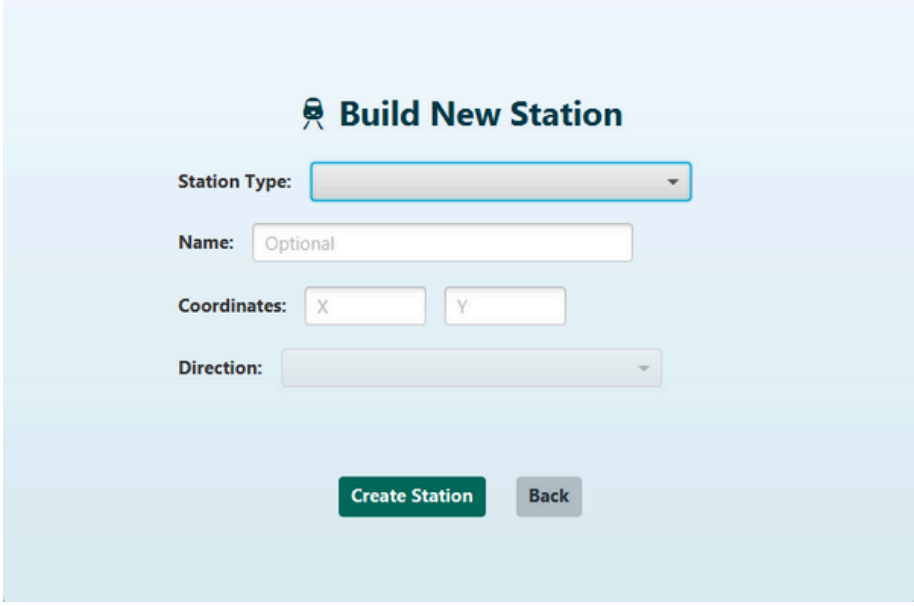


Figure 14 - Build Station Menu

Steps:

1. Select the **station type** from the dropdown menu (Depot, Station, Terminal).
2. *(Optional)* Enter a name for the station.
3. Enter the **X and Y coordinates** for the station placement.
4. *(Only for Station)* Select the direction
5. Click "**Create Station**" to confirm creation.

Validation Rules

Before building a station, the following validation rules must be respected:

- **Name (Optional):**
  - If filled, must not contain special characters.
- **Coordinates:**
  - X and Y values must be within the map boundaries.
  - The position must not overlap with another element (city, industry, or station).
  - Only positive integers are accepted.
- **Budget:**
  - The player must have enough funds to build the selected station type.
    - Example prices: Depot (50k), Station (100k), Terminal (200k).

If any of the above conditions are not met, an error message will be displayed and the station will not be created.

6.3.2. Upgrade Station

If the Player clicks on the "**Upgrade Station**" option, it will enter the menu that allows installing a building upgrade (e.g., telegraph, café, post office, hotel) to an existing station. These upgrades improve passenger satisfaction, cargo handling or station capacity.

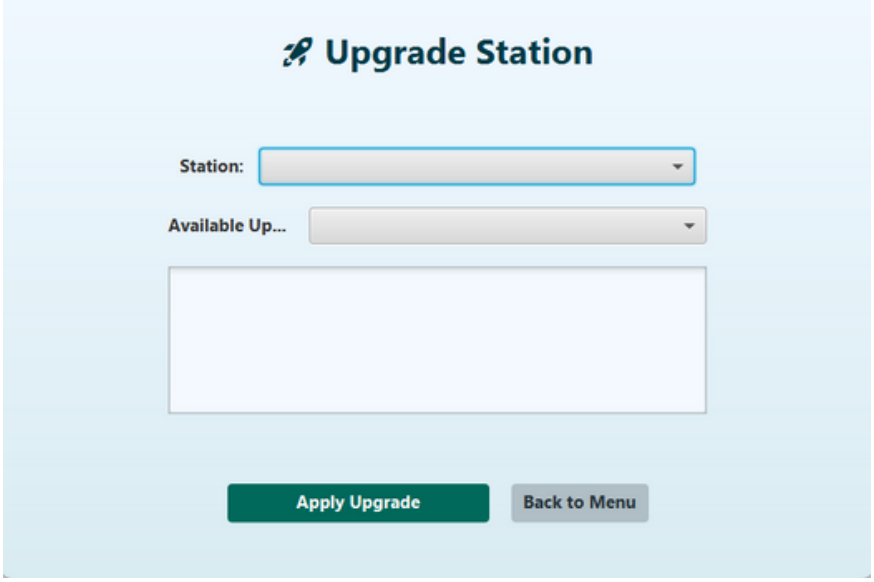


Figure 15 - Upgrade Station Menu

Steps:

1. Select the **station** to upgrade from the dropdown menu.
2. Choose one of the **available upgrades** compatible with that station and scenario year.
3. Review the **upgrade description**, including cost and effect, displayed in the text area.
4. Click "**Apply Upgrade**" to confirm installation.

Validation Rules

Before applying an upgrade, the following validation rules must be respected:

- **Available Upgrades:**
  - The list includes only buildings available in the current **scenario year**.
- **Upgrade Rules:**
  - A building that has already been installed on the station cannot be added again.
  - If the selected upgrade replaces another, the replaced building is removed automatically.
- **Budget:**
  - The player must have enough money to afford the selected upgrade.

If any of the above conditions are not met, an error message will be displayed and the upgrade will not be applied.

6.3.3. See Station Details

If the Player clicks on the "**See Station Details**" option, it will enter the menu that displays detailed information about an existing station on the map. This information helps the player assess the station's current configuration, performance, and potential for upgrades.

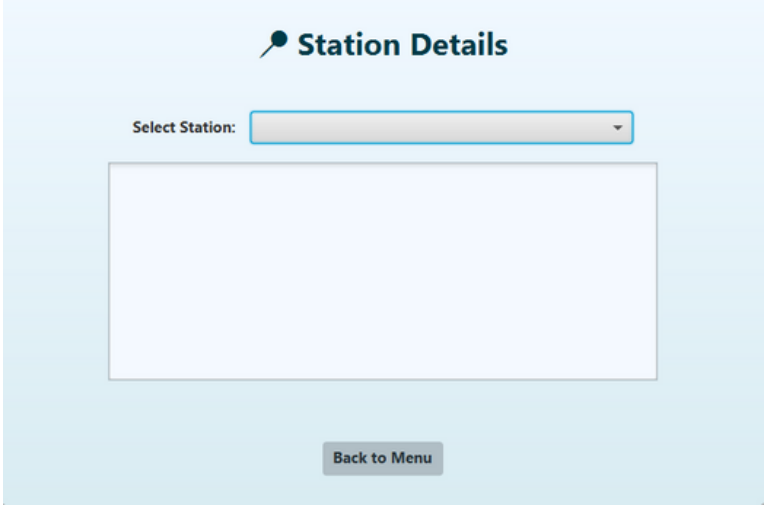


Figure 16 - See Station Details Menu

Steps:

1. Select a **station** from the dropdown menu.
2. *The system will display the station's full details in the text area below, including:*
   - Station name and type
   - Coordinates and direction
   - Installed buildings
   - Types of cargo supplied and demanded
3. Click "Back to Menu" to return to the main Player Menu.

Validation Rules

Before displaying the details, the following validation rules apply:

- **Station Selection:**
  - The station must be selected from the list of existing stations.

If no station is selected or if the station has no associated information, the text area will remain empty.

6.3.4. Build a Railway Line

If the Player clicks on the "**Build a Railway Line**" option, it will enter the menu that allows the creation of a railway line between two stations.



Figure 17 - Build a Railway Line Menu

Steps:

1. Enter the **name** of the railway line.
2. Select the **start station** from the dropdown list.
3. Select the **end station**.
4. Enter the **distance in kilometers** between the start and end stations.
5. *(Optional)* Add **intermediate stations** by selecting from the list and clicking "**Add Station**".
6. Click "**Build Line**" to confirm and create the railway connection.

Validation Rules

Before building the line, the following validation rules must be respected:

- **Line Name:**
  - Cannot be empty or null.
  - Must not contain special characters such as @, #, *, or /.
  - Must be unique (no duplicate names).
- **Station Selection:**
  - Start and end stations must be different.
  - Both stations must already exist on the map.
- **Distance:**
  - Must be a positive number.
- **Intermediate Stations (Optional):**
  - Each must be different from start and end stations.
- **Budget:**
  - The player must have enough funds to build the line (cost may depend on distance or type).

If any of the above conditions are not met, an error message will be displayed and the station will not be created.

6.3.5. Buy Locomotive

If the Player clicks on the "**Buy Locomotive**" option, it will enter the menu that allows purchasing a new locomotive. The locomotives are filtered by fuel type and availability according to the scenario's current year.

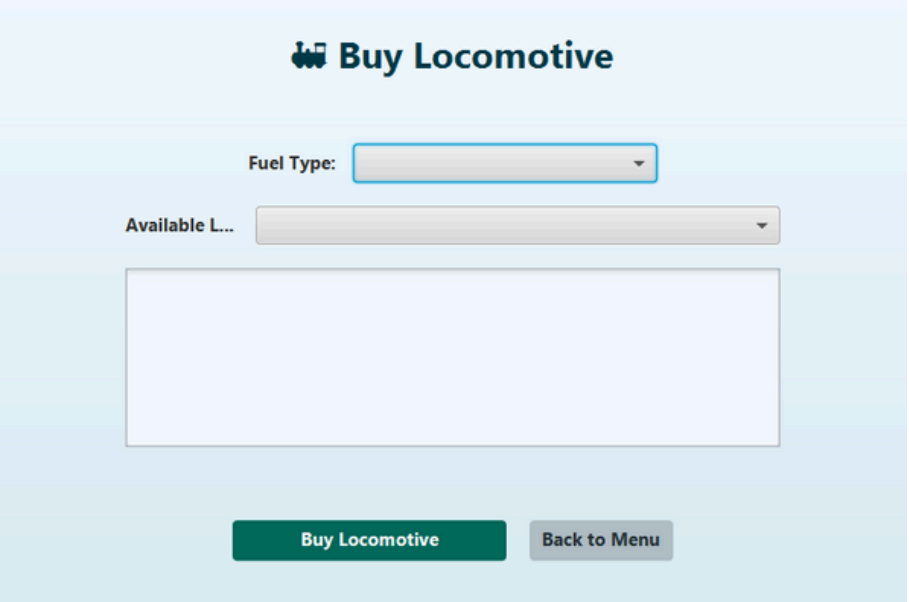


Figure 18 - Buy Locomotive Menu

Steps:

1. Select the **fuel type** from the dropdown (Steam, Diesel, Electric).
2. Choose one of the **available locomotives** compatible with the current year and selected fuel type.
3. Review the **locomotive's characteristics** (e.g., price, power, speed, cargo capacity) shown in the description area.
4. Click "**Buy Locomotive**" to confirm the purchase.

Validation Rules

Before buying a locomotive, the following validation rules must be respected:

- **Fuel Type:**
  - Must be selected from the list of types available in the current scenario.
- **Locomotive Selection:**
  - Only locomotives with availableFromYear ≤ current scenario year are listed.
  - The player must select one model before confirming.
- **Budget:**
  - The player must have sufficient funds to cover the locomotive's price.

If any of the above conditions are not met, an error message will be displayed and the locomotive will not be purchased.

6.3.6. Assign Train to Route

If the Player clicks on the "**Assign Train to Route**" option, it will enter the menu that allows assigning a route to a previously created train. The route is composed of multiple stations, each with a defined cargo strategy.

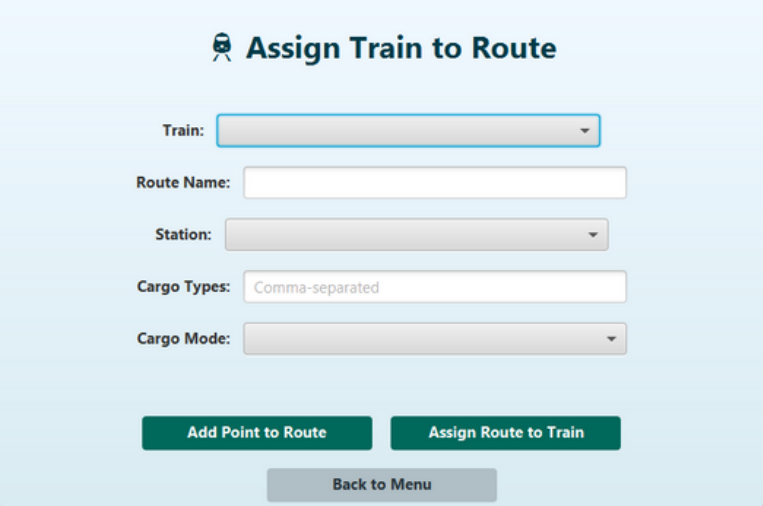


Figure 19 - Assign Train to Route Menu

Steps:

1. Select a **train** from the dropdown list of available trains.
2. Enter a **name** for the route.
3. For each stop on the route:
   - Select a **station** from the list.
   - Specify the **cargo types** (comma-separated, e.g., passengers, coal).
   - Select a **cargo mode**: FULL, HALF, or AVAILABLE.
   - Click "**Add Point to Route**" to register the stop.
4. After all route points have been added, click "**Assign Route to Train**" to confirm.

Validation Rules

Before creating a train, the following validation rules must be respected:

- **Train Selection:**
  - A train must be selected from the list of available trains.
  - Each train can only be assigned one route.
- **Route Name:**
  - Cannot be empty.
  - Must be unique.
- **Stations:**
  - At least two stations must be added (start and end).
  - Stations must be reachable and previously connected by railway lines.
  - No duplicate stations allowed in the same route.
- **Cargo Types:**
  - Must be valid and comma-separated.
- **Cargo Mode:**
  - Must be selected for each station.

If any of the above conditions are not met, an error message will be displayed and the route will not be assigned.

6.3.7. List all Trains

If the Player clicks on the "**List All Trains**" option, it will open a summary view of all trains created so far in the current simulation. This overview includes each train's ID, locomotive type, and assigned cargos.

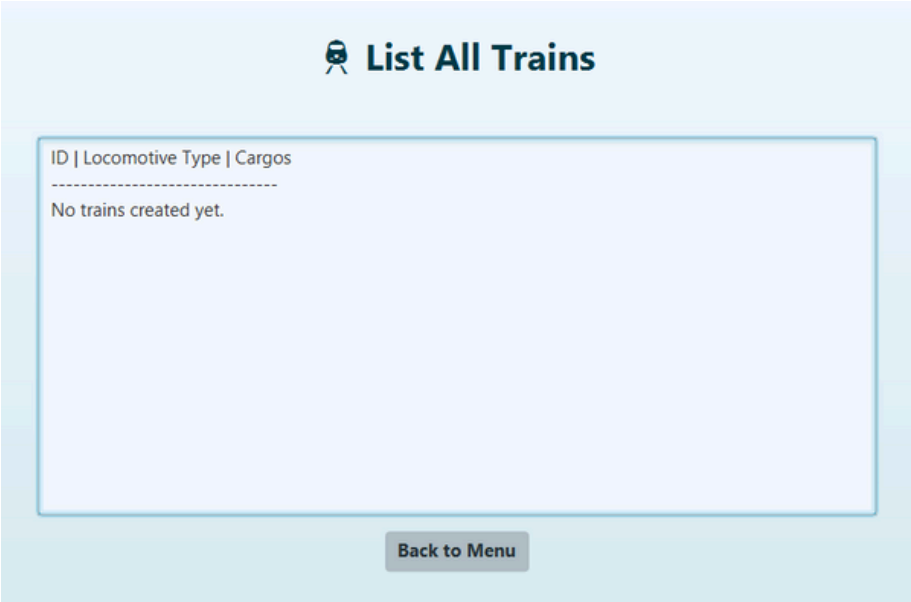


Figure 20 - List all Trains Menu

Steps:

1. Click "**List All Trains**" in the Player Menu.
2. The system will display a table-like summary with the following columns:
   - **ID:** Unique identifier for the train.
   - **Locomotive Type:** Type of locomotive used (e.g., Steam, Diesel, Electric).
   - **Cargos:** List of assigned cargo types.
4. If no trains exist, a message "**No trains created yet**." will be shown.
5. Click "**Back to Menu**" to return to the main screen.

Validation Rules

This feature is read-only, but the following constraints apply:

- **Data Display:**
  - Only trains that have been successfully created are shown.
  - Information must be up-to-date and reflect the current state of the simulation.
- **Formatting:**
  - Each train is listed on a separate line.
  - Data is presented in a consistent column format (ID | Locomotive Type | Cargos).

No user input is required. If no trains exist, the view remains empty with a default message.

7. TROUBLESHOOTING

This section outlines common technical issues encountered when running or using the Railway Network Simulator. It focuses on system-level and runtime problems that are not covered by functional validation rules.

Issue 1: Error loading map or scenario file

- **Symptom:** Message like "File not found" or "Could not load object".
- **Cause:** .ser file is missing, renamed, or corrupted.
- **Solution:** Ensure the file exists in the expected folder. Re-save from the Editor if needed.

Issue 2: Authentication file missing

- **Symptom:** Login screen fails to appear or gives "Authentication file not found".
- **Cause:** The system cannot locate the saved user data.
- **Solution:** Use the "Pick New Path" button to manually locate the file. If unavailable, contact the project supervisor.

Issue 3: Graphviz visualization does not appear

- **Symptom:** Railway network image not generated.
- **Cause:** Graphviz not installed or not configured correctly.
- **Solution:** Install Graphviz and ensure the dot executable is accessible via system PATH.

Issue 4: Simulation starts but trains don't move

- **Symptom:** Simulation clock runs, but nothing happens.
- **Cause:** No valid routes with compatible cargo deliveries.
- **Solution:** Verify that all trains have valid assigned routes with reachable stations and demand.

Issue 5: Application freezes during simulation

- **Symptom:** App becomes unresponsive after starting simulation.
- **Cause:** Map is too large or too many active entities (trains, industries).
- **Solution:** Reduce complexity of scenario. Recommended: use ≤ 10 trains and ≤ 30 stations for smooth performance on 8GB RAM systems.

Issue 6: Statistics tools not launching

- **Symptom:** Boxplots or regression graphs don't appear after simulation.
- **Cause:** Python or Jupyter Notebook not installed.
- **Solution:** Install Python 3

Issue 7: Low performance on older machines

- **Symptom:** Slow response, delays in UI, or long load times.
- **Cause:** Hardware below minimum spec.
- **Solution:** Use a system with at least:
  - Java 11+
  - JavaFX 11+
  - 8 GB RAM
  - Quad-core processor
  - SSD recommended

Issue 10: Platform compatibility

- **Symptom:** UI doesn't render or file paths break.
- **Cause:** Unsupported OS or 32-bit system.
- **Solution:** Use 64-bit versions of:
  - Windows 10/11
  - macOS 11+
  - Ubuntu 20.04+

ℹ️ **For in-game errors related to inputs (e.g., invalid coordinates, missing selections), please refer to the Validation Rules section of each feature.**

8. FREQUENTLY ASKED QUESTIONS (FAQS)

This section addresses common questions asked by users when interacting with the Railway Network Simulator. It is intended to clarify usage, improve usability, and prevent confusion about specific features or behaviors.

Q1: Can I build a station anywhere on the map?

Answer: No. You can only build a station on valid, unoccupied coordinates. The location must also respect the scenario rules (e.g., avoid overlapping cities or rivers).

Q2: Why are some buildings not available for upgrade?

Answer: Certain buildings (e.g., Telegraph, Café, Hotel) only become available in specific years, defined by the scenario. Additionally, only Station-type stations support upgrades — not Depots or Terminals.

Q3: I selected a locomotive fuel type, but the list is empty. Why?

Answer: The selected fuel type may not have any locomotives available for the current scenario year. Try choosing a different fuel type or verify the year settings of the scenario.

Q4: How do I assign a route to a train?

Answer: First, you must create a train by selecting a locomotive. Then, open the Assign Route option, choose the train, define the station sequence, and confirm the route. Only connected stations can be added.

Q5: What happens during the simulation?

Answer: The simulation generates cargo, moves trains along their routes, and calculates revenue based on deliveries. If no route is assigned or the cargo doesn't match industry demand, trains may move but no profit will be made.

Q6: Can I run multiple simulations or restart it?

Answer: Each scenario supports only one active simulation session. To restart, you must reload or create a new scenario.

Q7: Why can't I see the network map?

Answer: If using the visual graph feature, ensure Graphviz is installed and correctly configured. The visualization may also not appear if no stations or lines exist.

Q8: How do I know if a station has demand for a certain cargo?

Answer: Check the industries located within the station's range. Mixed or transforming industries often consume multiple cargo types. You can view this via the List Station Details option.

Q9: Can I remove or edit a station after creation?

Answer: No. Once a station is placed, it cannot be deleted or moved. You can only upgrade existing stations if they meet the required criteria.